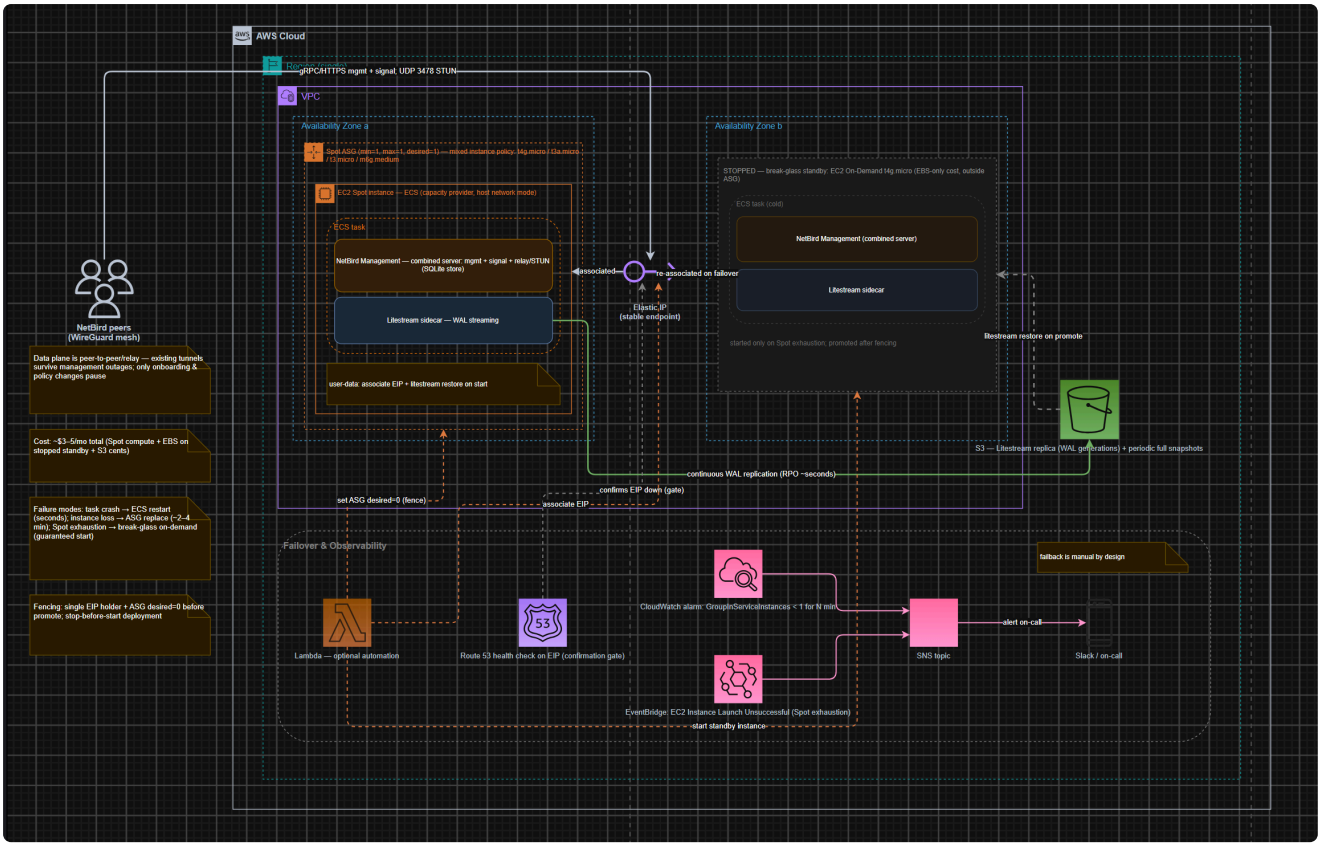
survey

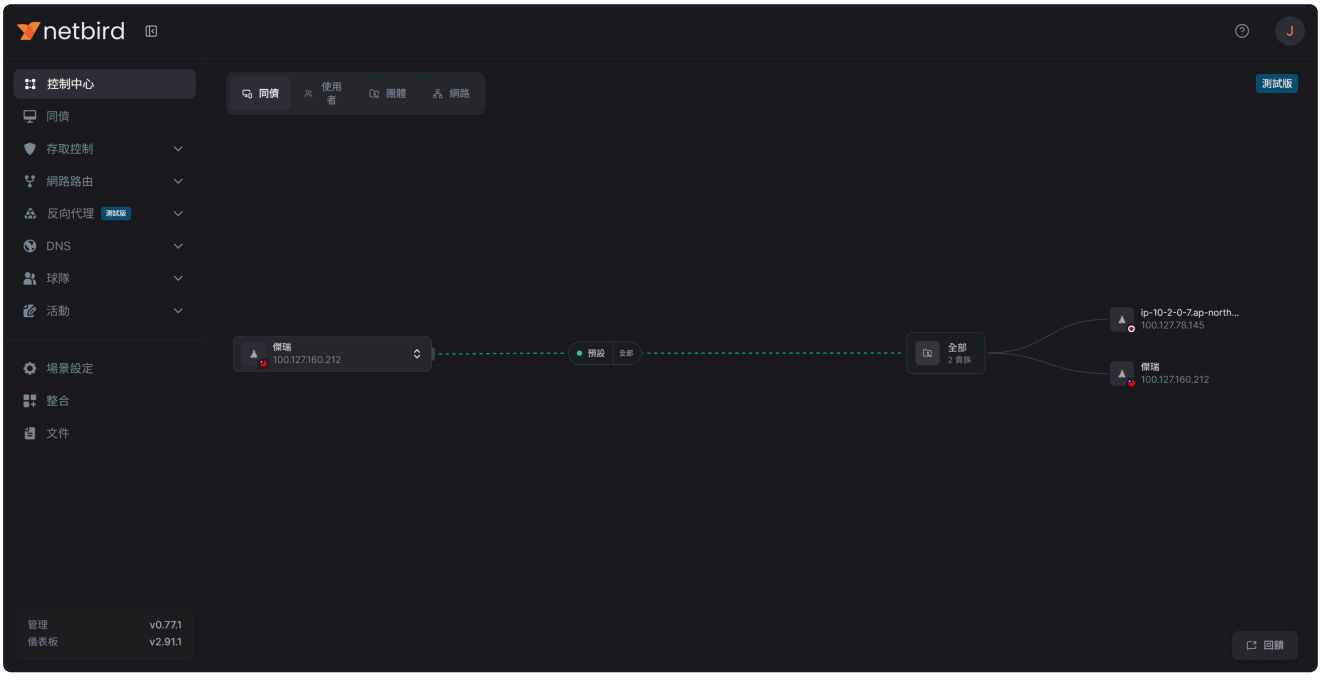
最一開始先使用一台 EC2 來研究

目標是把下圖的架構實作出來
我是在



但因為是要架設在 ecs 上面，我先用 ec2 來實驗比較方便

已經建好了，接下來是看要怎麼備份 netbird 的資料，先看他的架構大概是甚麼



看起來每台安裝 NetBird Client 的裝置成為 Peer 後，都會主動連線到 NetBird 的控制面(Management Service)，peer 第一次加入的時候會在本機產生 WG 的公私鑰，然後會將公鑰、電腦資料等等的資料送到

Management，然後 Management 會幫 peer 分配一個 100.64.0.0/10 內的虛擬 ip，並記錄它所屬的帳號、使用者、group、Access Policy、DNS、Network 與 router 設定，然後會一直維持連線

所以我要做的就只是直接把他的資料庫找出來並且將他們 mount 到同一個資料夾給 litestream 讀，但這是在 ec2 內的實驗環境，如果要在 ecs 裡面做的話，要先開一個 Volume 讓兩個 container 共用同一份資料

所以我可以先找到他預設設定的 compose 檔案並找到 netbird container 的名稱，再看他掛載的資料裡面可能就會有資料庫

從 docker compose.yaml 可以發現 netbird_data:/var/lib/netbird 這應該就是我要找的 Volume，所以需要將同一個 `netbird_data` Volume 也掛載到 Litestream Container 然後在 litestream.yml 裡面指定要備份的 sqlite 資料庫路徑跟目標 s3

```
services:
  # Traefik reverse proxy (automatic TLS via Let's Encrypt)
  traefik:
    image: traefik:v3.6
    container_name: netbird-traefik
    restart: unless-stopped
    networks:
      netbird:
        ipv4_address: 172.30.0.10
    command:
      # Logging
      - "--log.level=INFO"
      - "--accesslog=true"
      # Docker provider
      - "--providers.docker=true"
      - "--providers.docker.exposedbydefault=false"
      - "--providers.docker.network=netbird"
      # Entrypoints
      - "--entrypoints.web.address=:80"
      - "--entrypoints.websecure.address=:443"
      - "--entrypoints.websecure.allowACMEByPass=true"
      # Disable timeouts for long-lived gRPC streams
      - "--entrypoints.websecure.transport.respondingTimeouts.readTimeout=0"
      - "--entrypoints.websecure.transport.respondingTimeouts.writeTimeout=0"
      - "--entrypoints.websecure.transport.respondingTimeouts.idleTimeout=0"
      # HTTP to HTTPS redirect
      - "--entrypoints.web.http.redirections.entrypoint.to=websecure"
      - "--entrypoints.web.http.redirections.entrypoint.scheme=https"
      # Let's Encrypt ACME
      - "--certificatesresolvers.letsencrypt.acme.email=jerryfang6606@gmail.com"
      - "--certificatesresolvers.letsencrypt.acme.storage=/letsencrypt/acme.json"
      - "--certificatesresolvers.letsencrypt.acme.tlschallenge=true"
      # gRPC transport settings
      - "--serverstransport.forwardingtimeouts.responseheadertimeout=0s"
      - "--serverstransport.forwardingtimeouts.idleconntimeout=0s"

    ports:
      - '443:443'
      - '80:80'
    volumes:
      - /var/run/docker.sock:/var/run/docker.sock:ro
      - netbird_traefik_letsencrypt:/letsencrypt
```

```
    logging:
      driver: "json-file"
      options:
        max-size: "500m"
        max-file: "2"

  # UI dashboard
  dashboard:
    image: netbirdio/dashboard:latest
    container_name: netbird-dashboard
    restart: unless-stopped
    networks: [netbird]
    env_file:
      - ./dashboard.env
    labels:
      - traefik.enable=true
      - traefik.http.routers.netbird-
dashboard.rule=Host(`netbirdtest.5tmate.threatreveal.org`)
      - traefik.http.routers.netbird-dashboard.entrypoints=websecure
      - traefik.http.routers.netbird-dashboard.tls=true
      - traefik.http.routers.netbird-dashboard.tls.certresolver=letsencrypt
      - traefik.http.routers.netbird-dashboard.service=dashboard
      - traefik.http.routers.netbird-dashboard.priority=1
      - traefik.http.services.dashboard.loadbalancer.server.port=80
    logging:
      driver: "json-file"
      options:
        max-size: "500m"
        max-file: "2"

  # Combined server (Management + Signal + Relay + STUN)
  netbird-server:
    image: netbirdio/netbird-server:latest
    container_name: netbird-server
    restart: unless-stopped
    networks: [netbird]
    ports:
      - '3478:3478/udp'
    volumes:
      - netbird_data:/var/lib/netbird
      - ./config.yaml:/etc/netbird/config.yaml
    command: ["--config", "/etc/netbird/config.yaml"]
    labels:
      - traefik.enable=true
      # gRPC router (needs h2c backend for HTTP/2 cleartext)
      - traefik.http.routers.netbird-
grpc.rule=Host(`netbirdtest.5tmate.threatreveal.org`) &&
(PathPrefix(`/signalexchange.SignalExchange/`) ||
PathPrefix(`/management.ManagementService/`) ||
PathPrefix(`/management.ProxyService/`))
      - traefik.http.routers.netbird-grpc.entrypoints=websecure
      - traefik.http.routers.netbird-grpc.tls=true
      - traefik.http.routers.netbird-grpc.tls.certresolver=letsencrypt
      - traefik.http.routers.netbird-grpc.service=netbird-server-h2c
      - traefik.http.routers.netbird-grpc.priority=100
      # Backend router (relay, WebSocket, API, OAuth2)
```

```
      - traefik.http.routers.netbird-
backend.rule=Host(`netbirdtest.5tmate.threatreveal.org`) && (PathPrefix(`/relay`) ||
PathPrefix(`/ws-proxy/`) || PathPrefix(`/api`) || PathPrefix(`/oauth2`))
      - traefik.http.routers.netbird-backend.entrypoints=websecure
      - traefik.http.routers.netbird-backend.tls=true
      - traefik.http.routers.netbird-backend.tls.certresolver=letsencrypt
      - traefik.http.routers.netbird-backend.service=netbird-server
      - traefik.http.routers.netbird-backend.priority=100
      # Services
      - traefik.http.services.netbird-server.loadbalancer.server.port=80
      - traefik.http.services.netbird-server-h2c.loadbalancer.server.port=80
      - traefik.http.services.netbird-server-h2c.loadbalancer.server.scheme=h2c
    logging:
      driver: "json-file"
      options:
        max-size: "500m"
        max-file: "2"

volumes:
  netbird_data:
  netbird_traefik_letsencrypt:

networks:
  netbird:
    driver: bridge
    ipam:
      config:
        - subnet: 172.30.0.0/24
          gateway: 172.30.0.1
```

在最一開始我發現 Litestream 只對 Write-Ahead Logging (新資料先追加寫入 .db-wal) 的資料庫寫入方式有幫助， sqlite 的預設是 Rollback Journal (會先把資料備份到 .db-journal 再寫新資料進主要 db)，後來發現 Litestream 其實會在開始監控資料的時候(`litestream replicate` 時)，就把 sqlite 改成 WAL (`PRAGMA journal_mode = WAL;`)

將 litestream 接上之後，我嘗試將 db 刪掉，並試著還原，但我看不懂 litestream 是怎麼存資料的，雖然說只要一個指令就能還原啦...

小結

先用一台 ec2 把 netbird 架起來，然後看它的架構，確認資料存在哪裡。peer 第一次加入的時候在本機產生 WG 金鑰對，只把公鑰跟電腦資料送到 Management，Management 分配一個 100.64.0.0/10 內的虛擬 ip，並記錄它所屬的帳號、使用者、group、Access Policy、DNS、Network 與 router 設定。所以真正不能掉的資料都在 Management 那邊，備份要做的就是把 Management 的資料庫持續複製出去。

從 docker compose.yaml 找到 netbird_data:/var/lib/netbird 這個 Volume，ec2 上的做法很直接，把同一個 Volume 也掛給 litestream 的 container，在 litestream.yml 指定資料庫路徑跟 s3 目標就好。之後搬到 ecs 的話，等價的做法是開一個共用的 Volume 讓兩個 container 掛同一份資料。

中間繞了一下，litestream 只吃 WAL 模式的資料庫，sqlite 預設是 Rollback Journal，本來以為要自己改 journal_mode，後來發現 `litestream replicate` 開始監控的時候就會自己下 `PRAGMA journal_mode = WAL`，這件事不用管。

還沒解決的是還原。db 刪掉之後確實有一行指令救得回來，但還看不懂 litestream 在 s3 上是怎麼存的，這個還要再看。

參考資料

- How NetBird Works - NetBird Docs
- Advanced guide - NetBird Docs(self-hosted)
- Litestream
- How it works - Litestream
- Getting Started - Litestream(`litestream replicate` / `restore`)
- Write-Ahead Logging - SQLite
- PRAGMA journal_mode - SQLite
- Volumes | Docker Docs
- Storage options for Amazon ECS tasks(共用 volume)
- Traefik Docs
- RFC 6598 - IANA-Reserved IPv4 Prefix for Shared Address Space(100.64.0.0/10)